

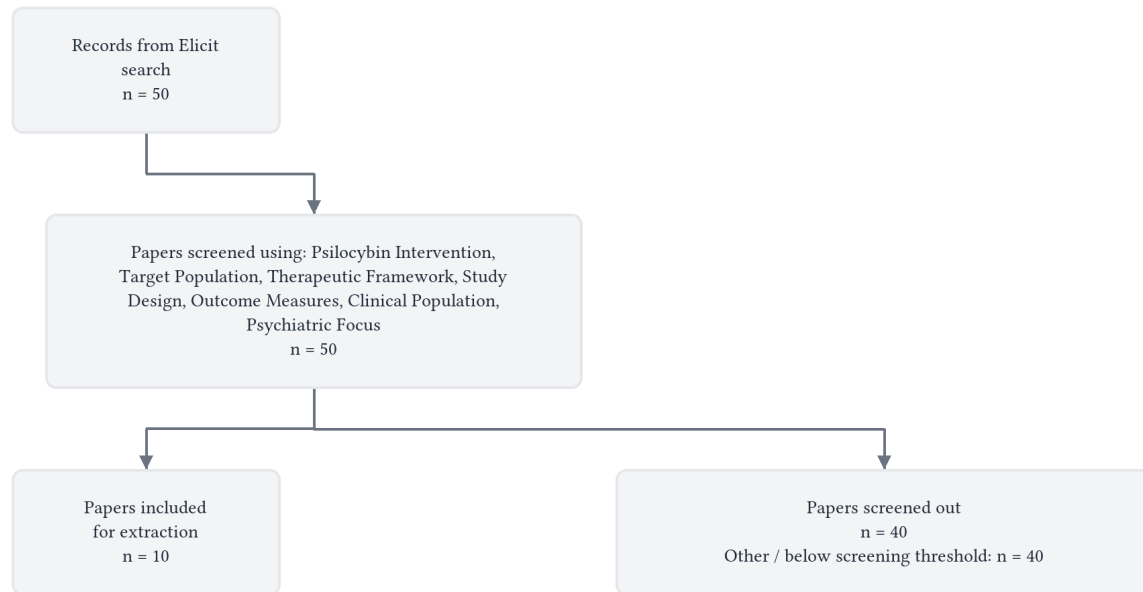
What are the differences in symptom reduction for depression, anxiety, or treatment-resistant conditions across Phase II trials of psilocybin-assisted therapy?

Symptom reduction in Phase II trials varied primarily by population, with life-threatening illness patients showing larger effects and better durability (60-80% maintained response at 6 months) compared to treatment-resistant depression populations (20% sustained response at 12 weeks), while 25 mg consistently emerged as the effective dose threshold.

Abstract

Phase II trials of psilocybin-assisted therapy demonstrated consistent symptom reduction for depression and anxiety, though the magnitude and durability varied substantially by population and protocol. Depression effect sizes ranged from $d=1.07$ to $d=2.97$ [1, 2], with the largest effects observed in patients with life-threatening illness ($d=2.97$ for depression, $d=4.51$ for anxiety) [2] and in open-label treatment-resistant depression studies ($d=2.2-2.3$) [3]. In the only direct comparison with an active antidepressant, psilocybin showed no significant difference from escitalopram on continuous depression scores ($p=0.17$) [4] but achieved higher response rates (70% vs 48%) and remission rates (57% vs 28%) [4]. Dose-response analysis indicated 25 mg as a threshold effective dose, with this dose significantly superior to 1 mg control ($p<.001$) while 10 mg showed intermediate, non-significant effects ($p=.18$) [5]. Treatment-resistant depression populations showed substantial acute responses that diminished over 3-6 months, with sustained response rates of only 20% at 12 weeks in one large trial [5], though repeated dosing protocols were associated with enhanced durability [1]. In contrast, patients with life-threatening illness maintained 60-80% response rates at 6.5 months [6] and sustained benefits at 26 weeks [2], suggesting this population experiences more durable effects. No serious adverse events occurred in most trials [1-4, 6, 7], though one trial reported suicidal ideation and self-injury events [5]. These findings indicate that psilocybin produces acute antidepressant and anxiolytic effects comparable to standard treatments, with effect durability and magnitude varying systematically by population context and chronicity of illness.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the differences in symptom reduction for depression, anxiety, or treatment-resistant conditions across Phase II trials of psilocybin-assisted therapy?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Psilocybin Intervention:** Does this study use psilocybin as the primary psychedelic intervention?
- **Target Population:** Does this study focus on participants diagnosed with depression, anxiety disorders, treatment-resistant depression, treatment-resistant anxiety, or other treatment-resistant psychiatric conditions?
- **Therapeutic Framework:** Does this study include a structured therapeutic component (such as preparation sessions, guided therapy sessions, and/or integration support) alongside psilocybin administration?
- **Study Design:** Is this study a Phase II clinical trial, controlled study with comparable methodology, or a systematic review/meta-analysis of such studies?

- **Outcome Measures:** Does this study report quantitative symptom reduction outcomes using validated measurement scales (e.g., MADRS, HAM-D, GAD-7, STAI, or similar validated instruments)?
- **Clinical Population:** Does this study focus on participants with clinical psychiatric conditions rather than healthy volunteers or consciousness research without therapeutic intent?
- **Psychiatric Focus:** Does this study investigate psychiatric/mental health applications rather than non-psychiatric conditions (such as cluster headaches or substance use disorders without comorbid target conditions)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 40

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design:**

Extract basic study characteristics including:

- Confirmation this is a Phase II trial
- Specific design (RCT, crossover, open-label extension, etc.)
- Study duration and follow-up periods
- Number of study sites
- Sample size (total enrolled and per group)

- **Target Conditions:**

Extract all specific psychiatric conditions studied in this trial, focusing on:

- Depression (specify type: major depressive disorder, cancer-related depression, life-threatening illness depression, treatment-resistant depression, etc.)
- Anxiety (specify type: generalized anxiety, cancer-related anxiety, existential anxiety, etc.)
- Any treatment-resistant conditions explicitly mentioned
- Comorbid conditions or dual diagnoses
- Inclusion criteria related to symptom severity or duration

- **Participant Characteristics:**

Extract key demographic and clinical features that may affect treatment response and generalizability:

- Age range and mean age
- Gender distribution
- Primary medical conditions (if applicable, e.g., cancer type, life-threatening illness)
- Baseline severity of depression/anxiety symptoms
- Previous treatment history or treatment resistance
- Any exclusion criteria that limit generalizability

- **Psilocybin Intervention:**

Extract complete details about the psilocybin treatment protocol:

- Dose(s) administered (mg, mg/kg, or other units)
- Number of psilocybin sessions
- Timing between doses (if multiple)
- Formulation (synthetic vs. natural)
- Setting and supervision during administration
- Duration of each dosing session
- **Psychological Support:**

Extract details about the therapy/psychological support component of psilocybin-assisted therapy:

- Type of psychological support (psychotherapy, counseling, etc.)
- Number of preparation sessions before psilocybin
- Number of integration sessions after psilocybin
- Duration and frequency of therapy sessions
- Therapist qualifications or training
- Any specific therapeutic modalities used
- **Control Groups:**

Extract information about comparator conditions:

- Type of control (placebo, active placebo, active comparator drug)
- Specific control substance and dose (if applicable)
- Whether control group received same psychological support
- Blinding procedures and effectiveness
- Any crossover design elements
- **Symptom Reduction Outcomes:**

Extract all symptom reduction results for depression, anxiety, and treatment-resistant conditions:

- Primary outcome measures used (MADRS, HADS, BDI-II, STAI, etc.)
- Baseline scores for each measure
- Post-treatment scores and change scores
- Effect sizes (Cohen's d, hedges g) where reported
- Statistical significance (p-values, confidence intervals)
- Response rates and remission rates (with definitions used)
- Separate results for depression vs. anxiety if both measured
- **Assessment Timepoints:**

Extract all time points when symptom reduction was assessed:

- Primary endpoint timing (e.g., 6 weeks, day 43)
- Secondary endpoint timings
- Long-term follow-up assessments
- Any immediate post-dose assessments
- Duration of sustained effects reported
- Any interim assessments between major timepoints

- **Safety Profile:**

Extract safety and tolerability data relevant to Phase II trials:

- Serious adverse events (number and description)
- Treatment-emergent adverse events by severity
- Dropout rates and reasons for discontinuation
- Any psychiatric adverse events (anxiety, paranoia, etc.)
- Medical adverse events
- Comparison of adverse event rates between treatment and control groups

- **Effect Durability:**

Extract information about the persistence of symptom reduction benefits:

- Longest follow-up timepoint with data
- Maintenance of symptom improvement at each follow-up
- Percentage of participants maintaining response/remission over time
- Any reported relapse rates
- Changes in effect size over time
- Any booster treatments or additional interventions during follow-up

Results

Characteristics of included studies

Study	Full text re-trieved?	Phase	Design	Condition	Sites	Sample Size	Psilocybin Dose(s)	Sessions	Control	Follow-up
R. Carhart-Harris et al., 2021	Yes	II [4]	Double-blind RCT [4]	Moderate-to-severe MDD [4]	Single site (implied) [4]	59 (30 psilocybin, 29 escitalopram) [4]	25 mg [4]	2 doses, 3 weeks apart [4]	Escitalopram (active comparator) [4]	6 weeks primary, 6 months planned [4]
C. Raison et al., 2023	Abstract only	II [7]	Double-blind RCT [7]	MDD ≥60 days, moderate or greater severity [7]	11 US sites [7]	104 (51 psilocybin, 53 niacin) [7]	25 mg synthetic [7]	Single dose [7]	100 mg niacin (active placebo) [7]	Day 43 [7]

Study	Full text re-trieved?	Phase	Design	Condition	Sites	Sample Size	Psilocybin Dose(s)	Sessions	Control	Follow-up
J. Rosenblat et al., 2024	Abstract only	Not stated	RCT with waitlist control [1]	TRD in MDD or bipolar II disorder [1]	Not mentioned [1]	30 (16 immediate, 14 delayed) [1]	25 mg [1]	1-3 sessions over 6 months [1]	Waitlist [1]	Week 2 primary endpoint, 6 months total [1]
Margaret Ross et al., 2025	Abstract only	Ib [2]	Double-blind RCT with open-label extension [2]	Depression with life-threatening illness [2]	Single site [2]	35 (17 psilocybin, 18 placebo) [2]	25 mg [2]	2 doses (after open-label extension) [2]	100 mg niacin (active placebo) [2]	26 weeks [2]
G. Goodwin et al., 2023	Abstract only	II [8]	Double-blind RCT [8]	TRD [8]	Not mentioned [8]	233 total [8]	COMP360 synthetic: 25 mg, 10 mg, 1 mg [8]	Single dose [8]	1 mg psilocybin [8]	Week 3 [8]
Stephen Ross et al., 2016	Yes	Not stated (likely II) [6]	Double-blind, placebo-controlled, and crossover [6]	Cancer-related anxiety and depression [6]	Single site (NYU) [6]	29 [6]	0.3 mg/kg synthetic [6]	Single dose (crossover at 7 weeks) [6]	250 mg niacin (active control) [6]	26 weeks (6.5 months) [6]
R. Carhart-Harris et al., 2017	Yes	Not stated [3]	Open-label [3]	Treatment-resistant major depression [3]	Single site (Imperial College) [3]	20 [3]	10 mg and 25 mg synthetic [3]	2 doses, 7 days apart [3]	None [3]	6 months [3]

Study	Full text re-trieved?	Phase	Design	Condition	Sites	Sample Size	Psilocybin Dose(s)	Sessions	Control	Follow-up
Otto Simons-son et al., 2023	Yes	Not applicable	Meta-analysis of 3 trials [9]	Depression [9]	Not mentioned [9]	102 (62 psilocybin, 29 escitalopram, 11 waitlist) [9]	Not specified [9]	Two doses [9]	Escitalopram and waitlist [9]	3 weeks [9]
W. Ishak et al., 2023	Abstract only	Not applicable	Systematic review of 5 RCTs [10]	Anxiety and depression [10]	Not mentioned [10]	Not mentioned [10]	14-30 mg/70kg (4 studies), 25 mg fixed (1 study) [10]	1-2 doses [10]	Not specified [10]	Up to 6 months [10]
G. Goodwin et al., 2022	Yes	II [5]	Double-blind, dose-finding, parallel-group RCT [5]	Treatment-resistant MDD [5]	Multiple sites (implied) [5]	233 (79 at 25 mg, 75 at 10 mg, 79 at 1 mg) [5]	COMP360 synthetic: 25 mg, 10 mg, 1 mg [5]	Single dose [5]	1 mg psilocybin [5]	12 weeks [5]

The included studies represent a mix of Phase II trials, open-label studies, and evidence syntheses examining psilocybin-assisted therapy for depression and anxiety. Six studies explicitly confirmed Phase II design [2, 4, 5, 7, 8]. Most studies targeted depression, with five specifically focusing on treatment-resistant depression (TRD) [1, 3, 5, 5, 8], two on cancer-related or life-threatening illness-associated depression and anxiety [2, 6], and two on major depressive disorder without treatment resistance specified [4, 7].

Sample sizes ranged from 20 [3] to 233 participants [5, 8]. Most trials used synthetic psilocybin [3, 5–7], with doses typically at 25 mg for higher-dose groups [1, 2, 4, 7], though one trial used weight-based dosing at 0.3 mg/kg [6]. Six trials employed active controls (niacin or escitalopram) [2, 4–7, 9], while others used low-dose psilocybin as control [5, 8] or waitlist designs [1]. One study was open-label without controls [3]. Follow-up periods ranged from 3 weeks [8] to 6 months [2, 3, 10].

Symptom reduction outcomes

Depression outcomes

Study	Measure	Baseline	Change Score	Effect Size	Statistical Significance	Response Rate	Remission Rate
R. Carhart-Harris et al., 2021	QIDS-SR-16	Psilocybin: 14.5 Escitalopram: 16.4 [4]	Psilocybin: - 8.0±1.0 Escitalopram: -6.0±1.0 [4]	Not reported	Between-group difference: 2.0 (95% CI -5.0 to 0.9), p=0.17 [4]	Psilocybin: 70% Escitalopram: 48% [4]	Psilocybin: 57% Escitalopram: 28% [4]
C. Raison et al., 2023	MADRS	Not reported [7]	Day 43: -12.3 Day 8: -12.0 [7]	Not reported [7]	p<.001 for both timepoints [7]	More in psilocybin (sustained response) [7]	Not significantly different [7]
J. Rosenblatt et al., 2024	MADRS	Not reported [1]	Greater reductions vs. waitlist [1]	Hedge's g=1.07 [1]	p<0.01 [1]	Not reported [1]	Not reported [1]
Margaret Ross et al., 2025	HADS-Depression	Not reported [2]	B=-2.49 [2]	d=1.12 [2]	p=.02 [2]	Not reported [2]	Not reported [2]
Margaret Ross et al., 2025	BDI-II	Not reported [2]	B=-7.56 [2]	d=2.97 [2]	p=.004 [2]	Not reported [2]	Not reported [2]
Stephen Ross et al., 2016	HADS, BDI	Not reported [6]	Significant improvements [6]	Large (Cohen's d) [6]	Significant [6]	60-80% at 6.5 months [6]	Significant rates [6]
R. Carhart-Harris et al., 2017	QIDS-SR16	Not reported [3]	Significant reductions [3]	Week 1: d=2.2 Week 5: d=2.33 months: d=1.56 months: d=1.4 [3]	p<.001 at all timepoints [3]	Week 5: 9/20 [3]	Week 5: 4/20 [3]
R. Carhart-Harris et al., 2017	BDI	Not reported [3]	Week 1: -22.73 months: -15.36 months: -14.9 [3]	Not reported [3]	p<.001 [3]	Not reported [3]	Not reported [3]

Study	Measure	Baseline	Change Score	Effect Size	Statistical Significance	Response Rate	Remission Rate
Otto Simonsson et al., 2023	Standardized depression measures	Not reported [9]	Not reported [9]	SMD: Psilocybin=-2.38 Escitalopram=1.56 [9]	Psilocybin vs. waitlist: OR=13.30, p=.001 Psilocybin vs. escitalopram: OR=0.88, p=.865 [9]	Not reported [9]	Not reported [9]
G. Goodwin et al., 2022	MADRS	All groups: ~32-33 [5]	25 mg: -12.0 10 mg: -7.91 5 mg: -5.4 [5]	Not reported [5]	25 mg vs. 1 mg: -6.6 (95% CI -10.2 to -2.9), p<.001 10 mg vs. 1 mg: -2.5 (95% CI -6.2 to 1.2), p=.18 [5]	Week 3: 25 mg: 37% 10 mg: 19% 5 mg: 18% [5]	Week 3: 25 mg: 29% 10 mg: 9% 5 mg: 8% [5]

Depression symptom reduction was consistently observed across studies, though the magnitude and durability varied. The largest effect sizes for depression were observed in the open-label trial by Carhart-Harris et al. (2017), with Cohen's $d=2.3$ at week 5 [3], and in the life-threatening illness trial by Ross et al. (2025), with $d=2.97$ for BDI-II [2]. The meta-analysis by Simonsson et al. (2023) found standardized mean differences of -2.38 for psilocybin [9].

In the direct comparison trial, Carhart-Harris et al. (2021) found no significant difference between psilocybin and escitalopram on the primary QIDS-SR-16 outcome at week 6 ($p=0.17$) [4], though response rates favored psilocybin (70% vs. 48%) [4] and remission rates were substantially higher for psilocybin (57% vs. 28%) [4]. The Raison et al. (2023) trial demonstrated significant MADRS reductions of -12.3 points at day 43 compared to niacin placebo ($p<.001$) [7].

Dose-response relationships were examined in the Goodwin et al. (2022) trial, which found 25 mg psilocybin significantly superior to 1 mg control (-6.6 point difference, $p<.001$) [5], while 10 mg showed intermediate effects that did not reach significance versus 1 mg control ($p=.18$) [5]. Response rates at week 3 were 37% for 25 mg, 19% for 10 mg, and 18% for 1 mg [5], with remission rates of 29%, 9%, and 8% respectively [5].

Anxiety outcomes

Study	Measure	Change Score	Effect Size	Statistical Significance
Margaret Ross et al., 2025	STAI-State	B=-12.59 [2]	$d=4.51$ [2]	$p=.005$ [2]

Study	Measure	Change Score	Effect Size	Statistical Significance
Stephen Ross et al., 2016	STAI, HADS-Anxiety	Significant improvements [6]	Large (Cohen's d) [6]	Significant [6]
R. Carhart-Harris et al., 2017	STAI-T	Week 1: -23.83 months: -12.26 months: -14.8 [3]	Not reported [3]	p<.001 [3]

Anxiety outcomes were reported in fewer studies but showed substantial reductions where measured. The Ross et al. (2025) trial in patients with life-threatening illness found extremely large effects for state anxiety ($d=4.51$) [2], suggesting particularly strong anxiolytic effects in this population. The Ross et al. (2016) cancer-related anxiety study similarly demonstrated sustained reductions in anxiety at 6.5-month follow-up, with approximately 60-80% of participants maintaining clinically significant improvements [6]. Trait anxiety reductions in the Carhart-Harris et al. (2017) open-label TRD trial showed mean decreases of -23.8 points at week 1, though these diminished somewhat to -12.2 and -14.8 points at 3 and 6 months respectively [3].

Treatment-resistant depression outcomes

Five studies specifically examined treatment-resistant depression. The open-label trial by Carhart-Harris et al. (2017) in patients with a median of 4 failed medication trials [3] showed large effect sizes ($d=2.2$ to 2.3) in the first 5 weeks [3], with 9 of 20 patients meeting response criteria and 4 meeting remission criteria at week 5 [3]. The Rosenblatt et al. (2024) study in complex TRD, including bipolar II disorder and high baseline suicidality [1], demonstrated a large effect (Hedge's $g=1.07$, $p<0.01$) for immediate treatment versus waitlist [1], with repeated doses associated with further symptom reductions [1].

The largest TRD trial by Goodwin et al. (2022) included 233 patients who had failed 2-4 adequate treatment trials [5] and showed 25 mg psilocybin superior to 1 mg control [5], though sustained response at 12 weeks was only 20% in the 25 mg group [5]. The Goodwin et al. (2023) report of exploratory outcomes from the same trial confirmed improvements across multiple patient-reported measures at week 3 [8].

Durability of effects

Study	Longest Follow-up	Effect Maintenance	Response/Remission Maintenance	Relapse Rate
R. Carhart-Harris et al., 2021	6 weeks primary endpoint, 6 months planned [4]	Not fully collected [4]	Not reported [4]	Not reported [4]
C. Raison et al., 2023	Day 43 [7]	Significant reduction maintained [7]	More sustained responses in psilocybin [7]	Not reported [7]
Margaret Ross et al., 2025	26 weeks [2]	Benefits sustained [2]	Not reported [2]	Not reported [2]

Study	Longest Follow-up	Effect Maintenance	Response/Remission Maintenance	Relapse Rate
Stephen Ross et al., 2016	6.5 months [6]	Sustained benefits in anxiety, depression, existential distress [6]	60-80% maintained response [6]	Not reported [6]
R. Carhart-Harris et al., 2017	6 months [3]	Effect size decreased from d=2.3 (week 5) to d=1.4 (6 months) [3]	6/9 responders maintained response [3]	3/9 responders relapsed [3]
G. Goodwin et al., 2022	12 weeks [5]	Sustained response: 25 mg=20%, 10 mg=5%, 1 mg=10% [5]	Week 3 response rates higher than week 12 [5]	Not reported [5]

Effect durability varied substantially across studies. The longest follow-up periods were 6 months [3] to 6.5 months [6], with the cancer-related anxiety study showing the most durable effects, with 60-80% of participants maintaining clinically significant improvements at 6.5 months [6]. The Carhart-Harris et al. (2017) open-label TRD trial showed declining but still significant effect sizes over time, from d=2.3 at week 5 to d=1.4 at 6 months ($p<.001$) [3], with 6 of 9 initial responders maintaining response at 6 months [3] and 3 experiencing relapse [3].

In contrast, the Goodwin et al. (2022) dose-finding trial showed more limited durability, with sustained response rates at 12 weeks of only 20% for the 25 mg group, 5% for the 10 mg group, and 10% for the 1 mg control [5]. The Ross et al. (2025) trial in life-threatening illness patients demonstrated sustained benefits at 26 weeks [2], suggesting this population may experience more durable effects. The Rosenblat et al. (2024) study suggested that repeated dosing over 6 months was associated with further reductions in depression severity compared to baseline [1], though specific long-term maintenance data were not reported [1].

Safety profile

Study	Serious Adverse Events	Overall Adverse Events	Dropout Rate	Key Adverse Events
R. Carhart-Harris et al., 2021	None in either group [4]	Psilocybin: 87% Escitalopram: 83% [4]	Psilocybin: 3 (COVID/guessing) Escitalopram: 5 (AEs/COVID) [4]	Psilocybin: none Escitalopram: anxiety, dry mouth [4]
C. Raison et al., 2023	None [7]	Higher overall and severe AEs in psilocybin [7]	Not reported [7]	Not specified [7]
J. Rosenblat et al., 2024	None [1]	Transient [1]	Low (29/30 retained to week 2) [1]	Not specified [1]

Study	Serious Adverse Events	Overall Adverse Events	Dropout Rate	Key Adverse Events
Margaret Ross et al., 2025	None [2]	More mild-to-moderate AEs with psilocybin [2]	1 withdrew due to anxiety during dosing [2]	Anxiety during dosing [2]
Stephen Ross et al., 2016	None [6]	Most common: BP/HR elevations (76%), headaches (28%), nausea (14%) [6]	Not reported [6]	Transient anxiety (17%), transient psychotic-like symptoms (7%) [6]
R. Carhart-Harris et al., 2017	None [3]	Transient anxiety (n=15), headaches (n=8), nausea (n=5), paranoia (n=3) [3]	1 chose not to complete follow-up [3]	Transient anxiety, paranoia [3]
G. Goodwin et al., 2022	Suicidal ideation, self-injury, hospitalization [5]	Headache (24%), nausea (22%), dizziness/fatigue (6%) [5]	25 mg: 6% 10 mg: 12% 1 mg: 13% [5]	Suicidal ideation, self-injury [5]

No serious adverse events were reported in most trials [1–4, 6, 7], with the notable exception of the Goodwin et al. (2022) trial, which reported serious adverse events including suicidal ideation, intentional self-injury, and hospitalization for severe depression [5]. Common adverse events across studies included headache [3–6], nausea [3, 5, 6], and transient anxiety [3, 6]. The Simonsson et al. (2023) meta-analysis found that clinically significant symptom worsening occurred in approximately 10% of participants receiving psilocybin, compared to 63.6% in the waitlist condition [9].

Transient psychiatric adverse events were reported in several studies, including anxiety (17% in Ross et al., 2016) [6], paranoia (n=3 in Carhart-Harris et al., 2017) [3], and transient psychotic-like symptoms (7% in Ross et al., 2016) [6]. Most adverse events were described as transient and resolving without pharmacological intervention [3, 6]. Medical adverse events were generally mild, with the most common being non-clinically significant elevations in blood pressure and heart rate (76% in Ross et al., 2016) [6].

Synthesis

The heterogeneity in outcomes across these Phase II trials reflects meaningful differences in patient populations, treatment protocols, and study designs rather than contradictory findings. Several factors explain the varying magnitudes and durability of symptom reduction.

Population and context distinctions: Studies targeting anxiety and depression in life-threatening illness contexts showed the largest and most durable effects. Ross et al. (2025) found an extremely large effect for anxiety ($d=4.51$) [2] with benefits sustained at 26 weeks [2], while Ross et al. (2016) demonstrated 60-80% response maintenance at 6.5 months in cancer patients [6]. This contrasts with treatment-resistant depression populations without life-threatening illness, where the Goodwin et al. (2022) trial showed only 20% sustained response at 12 weeks for 25 mg psilocybin [5]. The existential dimensions of life-threatening illness may create a particularly receptive context

for the psychological insights facilitated by psilocybin, potentially explaining stronger and more durable effects in these populations.

Treatment resistance and chronicity: Outcomes varied systematically with level of treatment resistance. The open-label Carhart-Harris et al. (2017) trial in patients with a median of 4 failed medications [3] showed very large initial effects ($d=2.3$) [3] but declining effect sizes over 6 months ($d=1.4$) [3] with 33% relapse rate among initial responders [3]. The Goodwin et al. (2022) trial requiring 2-4 failed trials [5] found more modest response rates (37% at week 3 for 25 mg) [5] and limited durability [5]. However, the Rosenblat et al. (2024) study in complex TRD including bipolar II disorder and baseline suicidality [1] achieved large effects ($g=1.07$) [1], suggesting that repeated dosing protocols (1-3 sessions over 6 months) [1] may enhance outcomes in highly resistant populations.

Dose-response relationships: The Goodwin et al. (2022) dose-finding trial demonstrated clear dose-dependent effects, with 25 mg significantly superior to 1 mg control ($p<.001$) [5] while 10 mg showed intermediate, non-significant effects ($p=.18$) [5]. This suggests a threshold dose around 25 mg for achieving robust antidepressant effects in treatment-resistant populations. The weight-based dosing in Ross et al. (2016) at 0.3 mg/kg [6] produced sustained effects in cancer patients, while most other trials converged on fixed 25 mg dosing [1, 2, 4, 7, 8].

Comparator and study design effects: The direct comparison between psilocybin and escitalopram by Carhart-Harris et al. (2021) failed to show superiority on the primary continuous outcome ($p=0.17$) [4], yet showed substantially higher response (70% vs. 48%) [4] and remission rates (57% vs. 28%) [4]. The Simonsson et al. (2023) meta-analysis confirmed no difference in symptom worsening risk between psilocybin and escitalopram ($OR=0.88$, $p=.865$) [9], while psilocybin showed substantial superiority over waitlist ($OR=13.30$, $p=.001$) [9]. This pattern suggests psilocybin achieves comparable overall efficacy to standard antidepressants but may produce more categorical improvements (crossing response/remission thresholds) in a subset of patients.

The open-label design of Carhart-Harris et al. (2017) [3] likely inflated effect sizes ($d=2.2-2.3$) [3] compared to blinded trials. However, the durability of effects in this uncontrolled context ($d=1.4$ at 6 months) [3] validates that psilocybin's benefits extend beyond acute placebo effects. The use of low-dose psilocybin (1 mg) as control in trials by Goodwin et al. raises questions about blinding effectiveness [5, 8], though this approach may better isolate dose-dependent pharmacological effects from psychological support and expectancy.

Temporal dynamics and dosing protocols: The largest early effect sizes ($d=2.2-2.3$ at weeks 1-5) [3] consistently diminished over time ($d=1.4-1.5$ at 3-6 months) [3], suggesting that single or dual-dose protocols produce peak effects acutely that partially attenuate. The Rosenblat et al. (2024) finding that repeated doses over 6 months produced further reductions beyond initial improvements [1] indicates that flexible, repeated dosing may enhance durability. The very long-lasting effects in life-threatening illness populations [2, 6] may reflect that these patients undergo more fundamental shifts in perspective regarding mortality and meaning that are less subject to regression.

Psychological support integration: While all trials included psychological support [2, 4-7], few provided detailed protocols. The Ross et al. (2025) trial used 3 preparatory and 6 integration psychotherapy sessions [2], potentially contributing to sustained benefits at 26 weeks [2]. The absence of specific therapeutic modality descriptions across most studies limits ability to determine whether variations in psychological support explain outcome heterogeneity.

Given these distinctions, psilocybin-assisted therapy demonstrates consistent acute efficacy across diverse depression and anxiety presentations, with 25 mg appearing to be a threshold effective dose. For patients with life-threatening illness, effects are large and durable (6+ months). For treatment-resistant depression without life-threatening illness, single-dose protocols produce substantial acute effects that diminish over 3-6 months, with repeated dosing potentially enhancing durability. Comparisons with standard antidepressants show similar overall efficacy but potentially greater categorical improvements in symptoms.

References

1. Rosenblat J, Meshkat S, Doyle Z, et al (2024) Psilocybin-assisted psychotherapy for treatment resistant depression: A randomized clinical trial evaluating repeated doses of psilocybin. *i Medicina*. <https://doi.org/10.1016/j.medj.2024.01.005>
2. Ross M, Iyer R, Williams M, et al (2025) Psilocybin-assisted psychotherapy for depression and anxiety associated with life threatening illness: A phase 2b randomized controlled trial. *General Hospital Psychiatry*. <https://doi.org/10.1016/j.genhosppsych.2025.08.001>
3. Carhart-Harris R, Bolstridge M, Bolstridge M, et al (2017) Psilocybin with psychological support for treatment-resistant depression: six-month follow-up. *Psychopharmacology*. <https://doi.org/10.1007/s00213-017-4771-x>
4. Carhart-Harris R, Giribaldi B, Watts R, et al (2021) Trial of Psilocybin versus Escitalopram for Depression. *New England Journal of Medicine*. <https://doi.org/10.1056/NEJMoa2032994>
5. Goodwin G, Aaronson S, Alvarez O, et al (2022) Single-Dose Psilocybin for a Treatment-Resistant Episode of Major Depression. *New England Journal of Medicine*. <https://doi.org/10.1056/NEJMoa2206443>
6. Ross S, Bossis AP, Guss JR, et al (2016) Rapid and sustained symptom reduction following psilocybin treatment for anxiety and depression in patients with life-threatening cancer: a randomized controlled trial. *Journal of Psychopharmacology*. <https://doi.org/10.1177/0269881116675512>
7. Raison C, Sanacora G, Woolley J, et al (2023) Single-Dose Psilocybin Treatment for Major Depressive Disorder: A Randomized Clinical Trial. *Journal of the American Medical Association (JAMA)*. <https://doi.org/10.1001/jama.2023.14530>
8. Goodwin G, Aaronson S, Alvarez O, et al (2023) Single-dose psilocybin for a treatment-resistant episode of major depression: Impact on patient-reported depression severity, anxiety, function, and quality of life. *Journal of Affective Disorders*. <https://doi.org/10.1016/j.jad.2023.01.108>
9. Simonsson O, Carlbring P, Carhart-Harris R, et al (2023) Assessing the risk of symptom worsening in psilocybin-assisted therapy for depression: A systematic review and individual participant data meta-analysis. *Psychiatry Research*. <https://doi.org/10.1016/j.psychres.2023.115349>
10. Ishak W, Garcia P, Pearl R, et al (2023) The Impact of Psilocybin on Patients Experiencing Psychiatric Symptoms: A Systematic Review of Randomized Clinical Trials. *Innovations in Clinical Neuroscience*